Nishchal Sapkota

Machine Learning Scientist — Vision Foundation Models • Multimodal AI • Reinforcement Learning • Healthcare AI

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Education

University of Notre Dame (UND) Notre Dame, IN
Ph.D. *in Computer Science and Engineering* *12/2026*
M.S. *in Computer Science and Engineering* *08/2024*
Research Areas: ***Computer Vision, Self-supervised Learning, Foundation Models, Data-efficient Deep Learning Models, AI for Healthcare***

The University of Southern Mississippi (USM) Hattiesburg, MS
B.S. *in* ***Computer Science*** | ***B.S.*** *in* ***Mathematics*** *– Honors, summa cum laude (GPA: 3.91)* *08/2020*
Thesis: Probabilistic Analysis of Revenues in Online Games

Experiences

Amazon Sunnyvale, CA
Applied Scientist Intern *08/2026 – 11/2026*
- Developed **data-efficient online reinforcement learning** for text-to-image personalization using sparse, region-level feedback and spatial credit assignment, outperforming preference optimization and SFT baselines. [1]
- Built large-scale multimodal generation and editing pipelines for animation production workflows, including synthetic dataset generation and a VLM-as-a-judge framework for evaluating fine-grained image edits, and RL-based artist-personalization.

Mayo Clinic Rochester, MN
Computational Pathology and AI Intern *01/2025 – 07/2026*
- Developed **one of the largest pathology vision foundation models** with 300× less data and achieving up to 5% improvement over state-of-the-art methods through scaling-aware training and mid-training regularization to resolve scaling bottlenecks [2]
- Built large-scale healthcare AI systems using ***distributed training*** and scalable ***MLOps pipelines*** for clinical predictive modeling. [10]

IBM Durham, NC
Senior Data Science & AI PhD Intern *05/2025 – 08/2025*
- Developed **user-behavior and session-level predictive models** (N-gram Markov, Transformer, Mamba) on large-scale web clickstream data to identify high-intent users and predict conversion outcomes, achieving a 14% improvement over internal baselines.
- Deployed models to enable real-time agentic AI marketing interventions, recovering up to **10,000 lost conversion opportunities per quarter**, while collaborating cross-functionally to translate model insights into business strategy.

The University of Notre Dame Notre Dame, IN
Graduate Researcher *08/2020 – Present*
- **Long Video Segmentation** [3]: Designed a novel dual-memory video segmentation model to advance AI-assisted surgery, outperforming foundational models (SAM2/3) in public surgical benchmarks by up to 17%
- **Data-Efficient Models for Image/Volume Segmentation** [4][5][6]: Pioneered data-efficient 2D and 3D segmentation models with up to 11% improvement on medical image segmentation benchmarks.
- **Self-supervised Learning Algorithms for Image Models** [11][12][13]: Built self-supervised learning frameworks achieving up to 4% gains across medical image segmentation benchmarks.
- **Structure-Aware Learning and Implicit Neural Representations** [7][14]: Proposed multimodal and shape-aware learning methods using implicit neural representations to address label ambiguity and improve data efficiency by 30%.
- ***Collaborated with academic, clinical, and industry labs*** to develop AI-driven healthcare solutions and ***mentored 5+ students*** [7], resulting in multiple publications and successful industry and academic placements. [7][15][16][17][18][19]

The University of Southern Mississippi Hattiesburg, MS
Undergraduate Researcher *08/2017 – 05/2020*
- Analyzed online games using Markov Chain to ***maximize revenues*** for both players and the providers. [8]
- Introduced a novel ***dynamic food chain model*** for three species and analyzed its long-term behavior. [9]
- Predicted ***chemical compound toxicity*** using in-vitro computational methods and feature engineering.

Technical Skills

Programming: Python, R, C++, Bash, MATLAB, SQL
ML Frameworks & Tools: PyTorch, TensorFlow, Hugging Face, Weights & Biases, Docker
Reinforcement Learning & Generative AI: Online RL (NFT, GRPO), Preference Optimization, Diffusion Models, Flow Matching, Text-to-Image (T2I) Model Adaptation, Parameter-Efficient Fine-Tuning (LoRA), Synthetic Data Generation, VLM-as-a-Judge
Vision & Representation Learning: Vision Foundation Models, Self-supervised Learning, Multimodal Learning, Vision-Language Models, Data-efficient Architecture Design
Healthcare AI & Medical Imaging: Computational Pathology, Whole-Slide Image (WSI) Analysis & Evaluation, Long-form Surgical Video Segmentation, MRI/CT Image Segmentation, Medical Vision Transformers
ML Systems: Distributed Training, Large-scale Model Pretraining, Memory-Efficient Training, MLOps Pipelines

Publications

First Author

[1] FlowRETAIN: Online Personalization of Text-to-Image Models from Sparse Region-Level Artist Feedback — *Under Review*

[2] BM1B: Data-Efficient Scaling of Bone Marrow Foundation Models through Domain-Specific Self-Supervised Learning and Dense Morphology-Preserving Representation Learning — *MedFMB @ ECCV, 2026*

[3] Sparsely Supervised Surgical Video Segmentation with Reliable Asymmetric Dual Memory – MICCAI, 2026

[4] When Swin Transformer Meets KANs: An Improved Transformer Architecture for Medical Image Segmentation — *IEEE ISBI, 2026*

[5] Universal Conditional Networks for Cartilage Segmentation with Sparsely Annotated Data — *Nature Scientific Reports, 2025*

[6] ConUNETR: A Conditional Transformer Network for 3D Micro-CT Embryonic Cartilage Segmentation — *IEEE ISBI, 2024*

[7] SHMC-Net: A Mask-guided Feature Fusion Network for Sperm Head Morphology Classification — *IEEE ISBI, 2024*

[8] Probabilistic Analysis of Revenues in Online Games — *University of Southern Mississippi, 2020*

[9] Hunting Co-operation in the Middle Predator in Three Species Food Chain Model — *MAA Proceedings, 2020*

Co-authored

[10] Lymphovision: A lymphoma-specialized foundation model for histology-based lymphoma classification and subtyping — *Blood, 2025*

[11] A Point in the Right Direction: Vector Prediction for Spatially-aware Self-supervised Volumetric Rep. Learning — *IEEE ISBI, 2023*

[12] Keep Your Friends Close & Enemies Farther: Debiasing Contrastive Learning with Spatial Priors in 3D Images — *IEEE BIBM, 2022*

[13] Unsupervised Feature Clustering Improves Contrastive Representation Learning for Medical Image Segmentation — *IEEE BIBM, 2022*

[14] SwIPE: Efficient and Robust Medical Image Segmentation with Implicit Patch Embeddings — *MICCAI, 2023*

[15] H-CNN-ViT: A Hierarchical Gated Attention Multi-Branch Model for Bladder Cancer Recurrence Prediction — *IEEE BIBM, 2025*

[16] IHCSurv: Effective Immunohistochemistry Priors for Cancer Survival Analysis in Gigapixel Multi-stain WSIs — *MICCAI, 2024*

[17] Path-GPTOmic: A Balanced Multi-modal Learning Framework for Survival Outcome Prediction — *IEEE ISBI, 2024*

[18] Boosting Medical Image Classification with Segmentation Foundation Model — *IEEE ISBI, 2024*

[19] Embryonic Cranial Cartilage Defects in the Fgfr$^{3Y367C/+}$ Mouse Model of Achondroplasia — *Anatomical Record, 2023*

Scholarships, Grants, Honors, and Achievements

2024 IEEE International Symposium on Biomedical Imaging (ISBI2024) **Travel grant** ($800)	ISBI 2024
Graduate School **Professional Development Fund** ($1,250) and **Conference Presentation Grant** ($450)	UND 2024
CSE Select **Fellowship** Award (*1/40 incoming Ph.D students; yearly stipend worth* $40,000)	UND 2020-2025
Wright W. and Annie R. Cross **Endowment** ($10,500) and Danny R. Carter Endowed **Scholarship** ($4,000)	USM 2017-2020
Eagle SPUR **grant** for Undergraduate Research ($2,000) and Honors Keystone **Scholarship** ($2,000)	USM 2019
First Place, Mathematics Comprehensive Exam (MFT)	USM 2019
Second Runner Up: Best Undergraduate Paper	MAA Meeting 2019
Finalist, Integration Bee	MAA Meeting 2018
Nominated for College of Science and Technology's **Outstanding Sophomore Award**	USM 2017
Burner Science & Tech. **Scholarship** ($800), Wallace C. & Lynn L. Pye Endowed **Scholarship** ($800)	USM 2017

Teaching Experiences

The University of Notre Dame — Notre Dame, IN

Graduate Teaching Assistant — *08/2020 – 12/2025*

- **Design & Analysis of Algorithms**; **Complexity & Algorithms**; **Mobile App. Design**; **Discrete Mathematics**
- Prepared lecture slides, graded submissions, created answer keys, and held office hours.

STEM Project Leader | Warrior-Scholar Project — *06/2023, 06/2024*

- **Medical Image Analysis**: Designed and conducted a Bootcamp to prepare veterans for undergraduate research.
- **Introduction to Data Science**: Conducted a Bootcamp to prepare veterans for undergraduate coding classes.

Students Mentored

- **Haoyan Shi** (Undergraduate Intern, 2025) – *Published author, Currently a PhD student at University of Notre Dame*
- **Zihao Zhao** (Undergraduate Intern, 2024) – *Currently a PhD student at Rutgers*
- **Maria Jose Gomez** (High School Intern, 2024) – *Published author, applying to undergraduate programs*
- **Sirui Li** (Undergraduate Intern, 2023) – *Currently a PhD student at UCLA*
- **Santiago Rodriguez** (Undergraduate Intern, 2023) – *Now a Software Engineer at Apple*